

Foundation (Jalapeno)

- Q1.** What is the factored form of $x^2 - 4$?
(A) $(x - 2)(x + 2)$
(B) $x(x - 4)$
(C) $x(x + 4)$
(D) $(x - 1)(x + 1)$
(E) $(x - 3)(x + 3)$
- Q2.** A spaceship travels at $a^2 - b^2$ km/h. If $a = 5$ and $b = 3$, what is the speed of the spaceship in km/h?
(A) 4
(B) 16
(C) 25
(D) 10
(E) 20
- Q3.** What is the factored form of $49 - y^2$?
(A) $7(y - 7)$
(B) $7(y + 7)$
(C) $(7 - y)(7 + y)$
(D) $(y - 7)(y + 7)$
(E) $y(7 - y)$
- Q4.** The gravitational force between two planets is $F = G \cdot \frac{m_1 m_2}{r^2 - d^2}$. If $r = 10$ and $d = 6$, what is the value of $r^2 - d^2$?
(A) 16
(B) 64
(C) 46
(D) 64
(E) 100
- Q5.** The factored form of $x^2 - 9$ is
(A) $x(x - 9)$
(B) $(x - 3)(x + 3)$
(C) $x(x + 9)$
(D) $(x - 1)(x + 1)$
(E) None of the above
- Q6.** In a lab experiment, the force exerted on an object is given by $F = ma^2 - mb^2$. If $m = 2$, $a = 4$, and $b = 2$, what is the force exerted on the object?
(A) 8
(B) 12
(C) 16
(D) 24
(E) 32
- Q7.** The equation $x^2 - 16 = 0$ can be factored as
(A) $x(x - 16) = 0$
(B) $(x - 4)(x + 4) = 0$
(C) $x(x + 16) = 0$
(D) $(x - 2)(x + 2) = 0$
(E) $(x - 8)(x + 8) = 0$
- Q8.** What is the factored form of $25 - y^2$?
(A) $5(y - 5)$
(B) $5(y + 5)$
(C) $(5 - y)(5 + y)$
(D) $(y - 5)(y + 5)$
(E) $y(5 - y)$

Open Ended Questions (FRQ)

- Q9.** The distance between two galaxies is given by $d^2 - c^2$, where d is the distance between the centers of the galaxies and c is the speed of light. If $d = 15$ and $c = 9$, find the value of $d^2 - c^2$. Show your work.
- Q10.** A spacecraft is traveling at a speed of $v^2 - u^2$, where v is the velocity of the spacecraft and u is the velocity of the solar wind. If $v = 8$ and $u = 4$, find the speed of the spacecraft. Explain your reasoning.

Chef's Tips

- Tip 1: Check for difference of squares patterns in equations and expressions.
- Tip 2: Make sure to apply the formula correctly: $a^2 - b^2 = (a - b)(a + b)$.
- Tip 3: Use real-world examples to make the concept more relatable and interesting.